ECE 340 # Homework Assignment #7 **Fall 2026**

DUE: FRIDAY, OCTOBER 23RD, 2026

MAKE SURE YOUR **NAME** AND **NETID** ARE CLEARLY VISIBLE. SUBMIT ASSIGNMENTS AS A PDF (EITHER SCANNED OR TYPED). SHOW ALL WORK (PARTIAL CREDIT CANNOT BE GIVEN WITHOUT WORK). SHOW ALL UNITS. HOMEWORK MUST BE TURNED IN THROUGH GRADESCOPE. HOMEWORK SHOULD BE TURNED IN ON TIME BY 5:00PM ON THE DUE DATE.

Unless specified otherwise, for all questions in this HW, assume silicon at temperature = 300K (kT = 0.026 eV) and $n_i = 1.5 \times 10^{10} cm^{-3}$

1. P-N JUNCTION UNDER BIAS

Consider a p-n junction made of Si at 300K. On the p-side, $N_A = 10^{17}\,\text{cm}^{-3}$ and on the n-side, $N_D = 5\times10^{17}\,\text{cm}^{-3}$. Assume $D_n = D_p = 25\,\text{cm}^2/\text{s}$ and $\tau_n = \tau_p = 10\,\text{ns}$.

(A). Calculate n_p and p_n at equilibrium.

(B). Calculate the diffusion length.

(C). If a forward bias voltage of 0.5V is applied, calculate the Δn_p and Δp_n at x_{p0} and x_{n0}. (Assume the applied voltage is less than the flat band voltage.)

(D). Calculate the hole diffusion current density $J_p(x = x_{n0})$ at x_{n0}, and the electron diffusion current density $J_n(x = x_{p0})$ at x_{p0} in the forward bias condition of 0.5V.

2. CARRIER INJECTION IN P-N JUNCTION

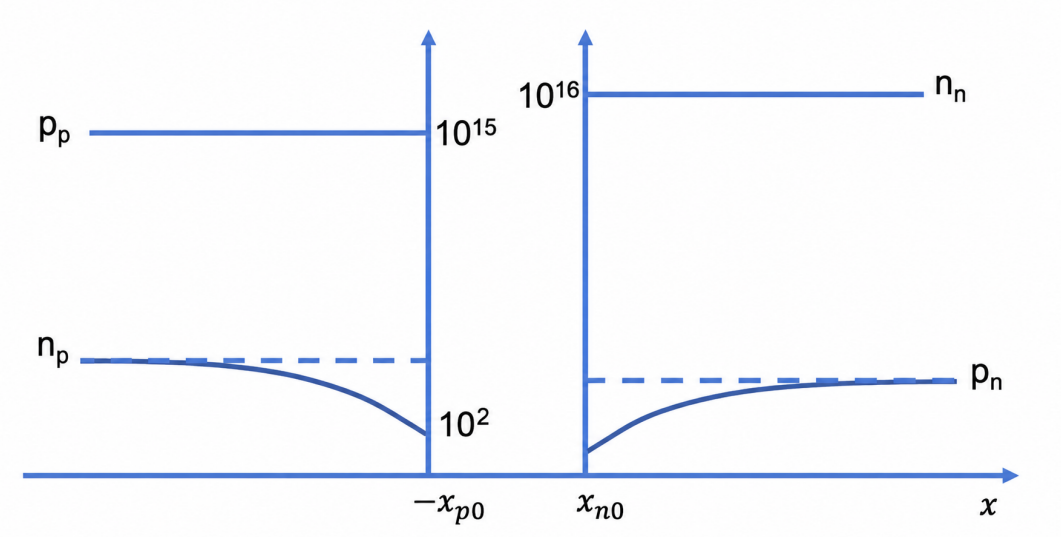
The figure below is a plot of the steady-state carrier concentration inside an abrupt p-n junction diode at room temperature. Assume $n_i = 1.5 \times 10^{10}\,\text{cm}^{-3}$.

(A). Is the diode forward or reverse biased? Explain how you arrived at your answer.

(B). Do low-level injection conditions prevail in the quasi-neutral regions of the diode? Explain why.

(C). What are the p and n side doping concentrations?

(D). Determine the applied voltage V_{bias}. What is the excess hole concentration in the n-side at x_{n0}?



3. Designing a Protection Diode

Your boss has tasked you with designing a p^+–n diode used for isolating a power supply from its load, used to stop reverse current from damaging the power supply. The customer gives you the specifications that at least $I = 1\,\text{mA}$ of current should flow when a forward voltage of $V_a = 0.7\,\text{V}$ is applied and that the cross-sectional area of the diode should be $A = 10^{-4}\,\text{cm}^2$.

The foundry you'll be fabricating this diode in uses Si and gives the following properties for its semiconductor technology:

$$D_p = 10\,\text{cm}^2/\text{s}, \qquad \tau_p = 100\,\text{ns}$$

Assume $n_i = 10^{10}\,\text{cm}^{-3}$ and $T = 300\,\text{K}$.

(A). This foundry allows for you to pick the doping amounts for your p and n region. However, for a p^+–n design, only one of these doping amounts matters, as long as the condition $N_A >> N_D$ is held. Using the diode equation, show which dopant (N_A or N_D) determines our diode's current using this approximation.

(B). Given your customers' requirements, now determine the amount of dopant we should use. You can just give the amount of dopant necessary for the dominant dopant found in part (a), and can assume the foundry dopes the other such that the condition $N_A >> N_D$ is held.

(C). Find the ratio $I(V_a = 0.8\,\text{V})/I(V_a = 0.7\,\text{V})$.

(D). Your customer attempts to use your diode in series between a 5V power supply and a load. The forward-biased diode quickly gets destroyed. Why did the diode break, and how would you fix this circuit? (*Hint: You only need a passive circuit element.*)

4. Junction Doping and Breakdown Behavior

Consider a silicon p^+–n junction, where the n-side doping concentration is doubled. Assume the geometry, temperature, and material properties remain constant, and that avalanche breakdown dominates initially. For each part below, indicate whether the quantity increases, decreases, or remains unchanged. Briefly explain your reasoning.

(A). How does the junction capacitance under zero bias change?

(B). How does the built-in potential V_{bi} change?

(C). How does the reverse-bias breakdown voltage change? Which mechanism (Zener or avalanche) becomes more likely?

(D). How does the temperature dependence of breakdown voltage change after the doping modification?